Mathematics Department

Grade 12 SE

Statistics in two variables

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Definition

We call statistics in two variables the study of two characteristics to the same population .

Example : we study the height x_i in cm and the weight y_i in kg of five students of a class

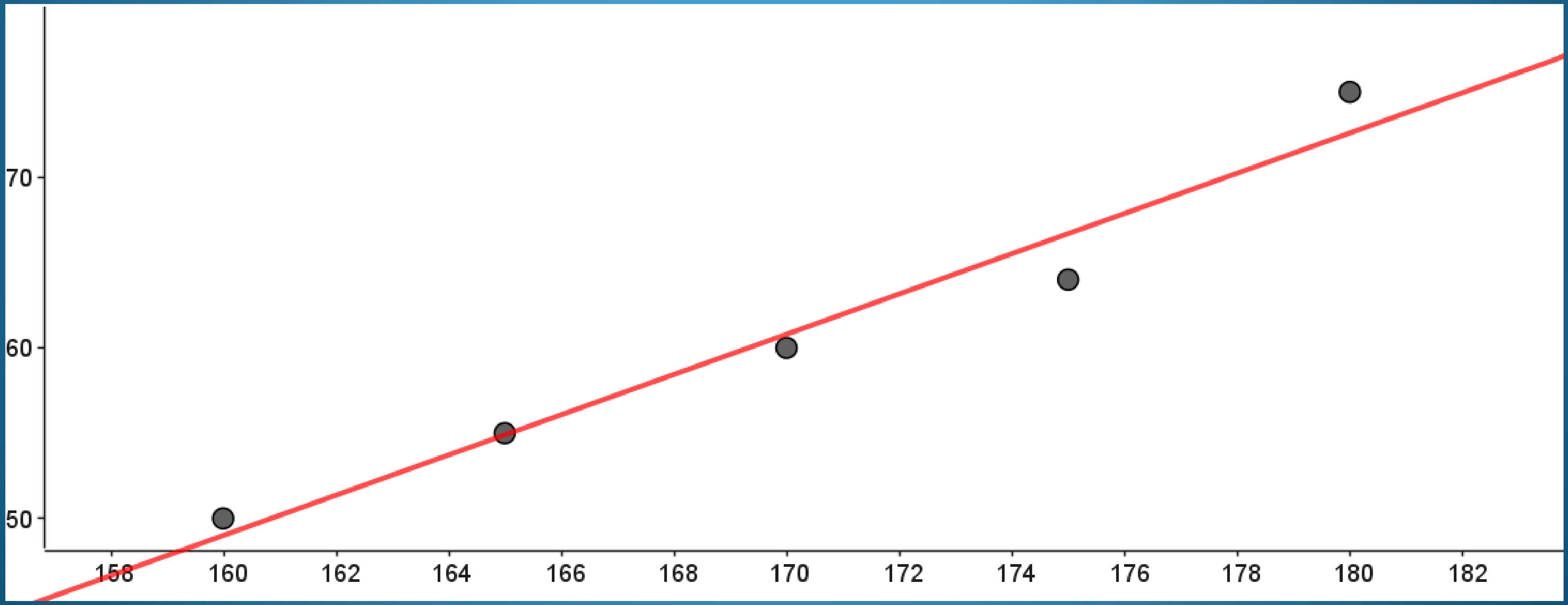
We can represent the data by the following table

Height x_i in cm	**160**	**165**	**170**	**175**	**180**
weight y_i in kg	50	55	60	64	75

Scatter plot

The set of points $M(x_i; y_i)$ is called scatter plot.

Example : The scatter plot of the above table can be represented as follows:



Sense of variation of Y in terms of X:

If Y increase as X increase. Then we say that there is a positive correlation between X & y.

If Y decrease as X increase then we say that there is a negative correlation between X & Y.

Mean point

The point $G(\bar{x}, \bar{y})$ is called the center of gravity or mean point where

$\bar{x} = \frac{\sum x_i}{n}$ & $\bar{y} = \frac{\sum y_i}{n}$ are the respective means of X & Y.

Example:

In the above example $\bar{x} = \frac{160+165+170+175+180}{5} = 170$

$$\bar{y} = \frac{50+55+60+64+75}{5} = 60.8$$

therefore the mean point is G(170,60.8)

variance and standard deviation

Variance of X: $V(x) = \frac{\sum {x_i}^2}{n} - \bar{x}^2$

Standard deviation of X: $\sigma(X) = \sqrt{V(X)}$

Variance of Y: $V(Y) = \frac{\sum {y_i}^2}{n} - \bar{y}^2$

Standard deviation of Y: $\sigma(Y) = \sqrt{V(Y)}$

Example :

In the above example

$$V(x) = \frac{\sum {x_i}^2}{n} - \bar{x}^2 = \frac{160^2+165^2+170^2+175^2+180^2}{5} - 170^2 = 50$$

$$\sigma(X) = \sqrt{V(X)} = \sqrt{50} = 5\sqrt{2} = 7.07$$

covariance

The covariance of the variables X & Y is given by:

$$\mathrm{cov}(X;Y) = \frac{\sum x_i y_i}{n} - \bar{x}\bar{y}$$

Example :

in the above example

$$\mathrm{cov}(X;Y) = \frac{160\times55+165\times55+170\times60+175\times64+180\times75}{5} - 170 \times 60.8$$

$$= 59$$

Linear adjustment – regression line:

In the case where a linear adjustment is justified, we determined the equation of the regression line ($D_{y/x}$), of y in x, by using the least squares method; y= Bx+A with;

$$B = \frac{\text{cov}(X;Y)}{V(X)} \text{ and } A = \bar{y} - B\bar{x}$$

There exist another regression line ($D'_{x/y}$), x in y, of equation x= B'y+A' where $B' = \frac{\text{coV}(X;Y)}{V(Y)}$ and $A' = \bar{x} - B'\bar{y}$

Remark: The regression line has to pass through the center of gravity G. Therefore the two regression lines (D) and (D') intersect at G.

Example

the regression line from y to x $(D_{y/x})$:y= Bx+A

$$B = \frac{\text{cov}(X;Y)}{V(X)} = \frac{59}{50} = 1.18$$

$$A = \bar{y} - B\bar{x} = 60.8 - 1.18 \times 170 = -139.8$$

Therefore $(D_{y/x})$:y= 1.18x-139.8

Estimated value of y knowing a value of x

Estimate the weight of a student knowing that its height is is 200 cm .

Answer :

we have y= 1.18x-139.8

if the height =200 cm then x=200

if x=200 then y=1.18 ×200-139.8=96.2

therefore the estimated weight for a student of height 200cm is 96.2 kg

% of error between the reality and the estimated value

in reality the weight of a student of height 200 cm is 90 kg

% error $=\frac{|R-E|}{R} \times 100 = \frac{|90-96.2|}{90} \times 100 = \frac{6.2}{90} \times 100 = 6.88\%$

The coefficient of correlation:

The coefficient of correlation of the series (X;Y) is given by:

$$r = \frac{\text{cov}(X;Y)}{\sigma(X) \times \sigma(Y)}$$

Example :

In the above example $r = \frac{\text{cov}(X;Y)}{\sigma(X) \times \sigma(Y)} = \frac{59}{7.07 \times 8.51} = 0.98$

Interpretation of the value of r

If $0 < r < 0.5$ then there is a weak positive linear correlation between X & Y.

If $0.5 < r < 0.75$ then there is an average positive linear correlation between X & Y.

If $0.75 < r < 1$ then there is a strong positive linear correlation between X & Y.

If $-0.5 < r < 0$ then there is a weak negative linear correlation between X & Y.

If $-0.75 < r < -0.5$ then there is an average negative linear correlation between X & Y.

If $-1 < r < -0.75$ then there is a strong negative linear correlation between X & Y.

Important percentages:

Percentage of increase of a variable: $\frac{F-I}{I} \times 100$

Percentage of decrease of a variable: $\frac{I-F}{I} \times 100$

Percentage of error in the estimation of a variable:

$\frac{|R-E|}{R} \times 100$

With:

F= the final value of the variable.

I= the initial value of the variable.

R= the real value of the variable.

E= the estimated value of the variable.